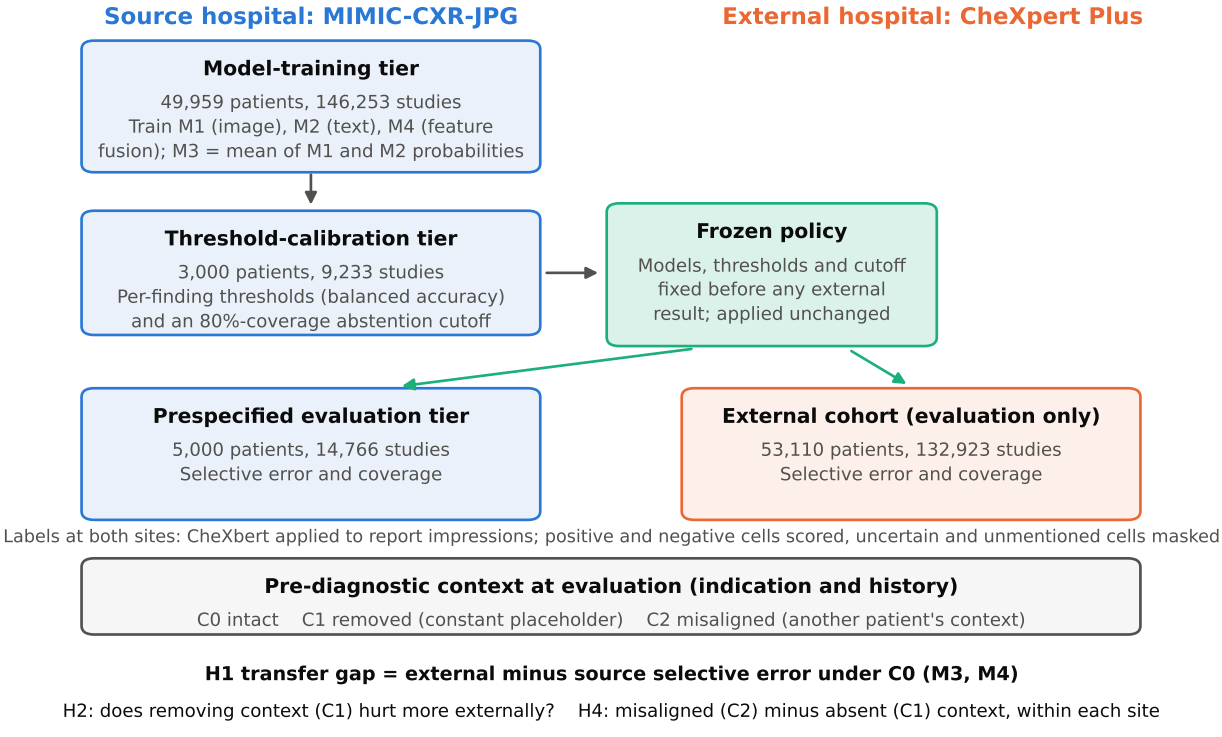
Source hospital: MIMIC-CXR-JPG
External hospital: CheXpert Plus
Model-training tier
49,959 patients, 146,253 studies
Train M1 (image), M2 (text), M4 (feature fusion); M3 = mean of M1 and M2 probabilities
Threshold-calibration tier
3,000 patients, 9,233 studies
Per-finding thresholds (balanced accuracy) and an 80%-coverage abstention cutoff
Frozen policy
Models, thresholds and cutoff fixed before any external result; applied unchanged
Prespecified evaluation tier
5,000 patients, 14,766 studies
Selective error and coverage
External cohort (evaluation only)
53,110 patients, 132,923 studies
Selective error and coverage
Labels at both sites: CheXbert applied to report impressions; positive and negative cells scored, uncertain and unmentioned cells masked
Pre-diagnostic context at evaluation (indication and history)
C0 intact C1 removed (constant placeholder) C2 misaligned (another patient's context)
H1 transfer gap = external minus source selective error under C0 (M3, M4)
H2: does removing context (C1) hurt more externally? H4: misaligned (C2) minus absent (C1) context, within each site